Discrimination of Green Tea Samples on the Basis of Gallic Acid Content Using Near Infrared Spectroscopy

Debangana Das[1, *], Shreya Nag[1], Hemanta Naskar[1], Tanmay Sau[1], Ajanto Kumar Hazarika[2], Santanu Sabhapondit[2], Runu Banerjee Roy[1] , Bipan Tudu[1] and Rajib Bandyopadhyay[1,3]

[1]Department of Instrumentation and Electronics Engineering, Jadavpur University, Kolkata, India
*Email: debanganadas4@gmail.com
[2] Tocklai Tea Research Institute, Jorhat, Assam
[3]Laboratory of Artificial Sensory Systems, ITMO Univerity, Saint Petersberg, Russia

Abstract

Green tea is crowned as a wonder beverage and has hogged the imagination of a huge population due to its anti-oxidant properties. In this work, the classification of five variants of green tea has been done on the basis of its gallic acid content which is a major anti-oxidant in green tea using NIR spectroscopy. The reflectance spectra of the green tea samples were recorded in the wavelength range of 900 - 1700 nm. The gallic acid contents in the tea samples were derived using High Performance Liquid Chromatography (HPLC). These HPLC values were used as the reference values for partial least square regression (PLSR) model development using NIR spectral data. Initially, principal component analysis (PCA) has been applied to study the discrimination capabability of the NIR spectral data. The well defined clusters have been formed in the PCA plot. For quantitative evaluation of the data nature, class separability index (SI) has been calculated. SI value obtained for raw dataset was 53.63. Different data pre processing techniques were applied on the raw dataset and the SI value increased to 93.20. Hence, the results demonstrate that NIR spectroscopy coupled with suitable chemometrics can be applied successfully as a rapid method for the discrimination of green tea samples. This application can be used for discrimination of other food stuffs as well.

Keywords: green tea, gallic acid, principal component analysis, partial least square regression, HPLC.

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